

CMPUT 654: Mathematical Foundations of Modern AI Systems

Fall 2026

Lecture 5: State-space models and system identification

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About these lecture notes. This lecture asks how a structured hidden state can explain a sequence, how to predict and generate after observing a prefix, and how to fit the model from observations. The main construction first fits observed lag coefficients, then recovers a compact recurrent predictor by matrix factorization.

Scope and reading guide. The main text follows the classroom sequence: nonlinear state-space models, conditional generation, estimation and identifiability, latent likelihoods and EM, observable prediction, linear models, and the subspace algorithm. It develops the questions about memory, parameter sharing, backpropagation, and attention raised in class. The synthetic comparisons are additions to the lecture. The optional section after this development justifies the algorithm and relates estimation error to prediction, sequence KL, and generation. Cayley–Hamilton was not presented in class; it is retained as an optional noiseless comparison and an exercise. After the take-home messages, the notes continue with endnotes, bibliographical notes and references, the glossary, exercises, and appendices.

Notation. States are $S_t \in \mathcal{S}$, observations are $Z_t \in \mathcal{Z}$, and $Z_{<t} = Z_{1:t-1}$. Finite observation labels are denoted X_t . Vectors are columns. T is trajectory length; d and m are state and observation dimensions. Expectations refer to the model law specified at their use; logarithms are natural. Lecture 4 supplies the sequence KL and recovery background. Superscript asterisks* mark optional material; numbered endnotes contain short refinements.

5.1 State, observations, and the modeling choice

A state-space model specifies what information a process retains and how that information produces observations. Let $\mathcal{S}$ and $\mathcal{Z}$ be the state and observation spaces. We use the equations

$$S_0 \sim \pi_\theta, \quad S_{t+1} = f_\theta(S_t, V_{t+1}), \quad Z_{t+1} = g_\theta(S_{t+1}, W_{t+1}), \quad t \geq 0. \quad (5.1)$$

Here $(V_t)_{t \geq 1}$ and $(W_t)_{t \geq 1}$ are independent iid sequences, independent also of S_0 . Their laws are part of the model and may depend on θ . The two noise laws need not be the same. We assume measurable update maps; Euclidean spaces and finite sets cover our examples.¹

The process (S_t) is Markov: conditional on its current state, its next state does not depend on earlier states or observations. We call the full construction a *hidden Markov model* (HMM), and (Z_t) its *observation process*, or a *hidden Markov process*. The pair process $(S_t, Z_t)_{t \geq 1}$ is Markov. The observation process alone generally is not. We will say *finite HMM* when $\mathcal{S}$ is finite. This terminology also allows a continuous-state HMM.

5.1.1 Where the inductive bias enters

The useful assumption is a small or structured state space and a restricted update rule. As in Lecture 1, an architecture relates predictions at different inputs through shared structure. For

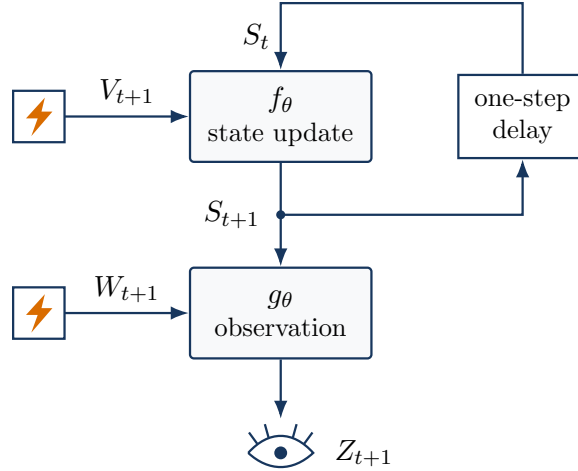


Figure 1: A state-space model. Independent noise sources enter the update and observation maps. The state feeds both the observation map and the next update; the one-step delay is initialized with S_0 . Dots mark branches of the same signal. The eye marks the observed output.

physical motion, a few state coordinates can have known meanings. For language, identifying a useful compact state is itself a learning problem.

With unrestricted state space and maps, the state can record the entire history. More precisely, it can store a finite observation string, sample its next symbol from the desired conditional law, append that symbol, and emit it. The string includes its length, so this construction can also describe a process whose conditional law changes with time. It gives a state-space representation of any process with the required conditional distributions. The growing state supplies no compression or structural restriction.

The representation is a modeling choice. Removing process noise gives

$$S_{t+1} = f_{\theta}(S_t), \quad Z_t = g_{\theta}(S_t, W_t).$$

All state randomness then comes from the initial state. Another choice computes a predictor's state from observed feedback:

$$R_{t+1} = h_{\theta}(R_t, Z_t), \quad Z_t = j_{\theta}(R_t, W_t), \quad R_1 = r_{\theta}. \quad (5.2)$$

Given a prefix, R_t is known. During generation, random observations feed randomness into future states.

The pair (R_t, Z_t) is Markov, but observing Z_t need not reveal R_t . Only an additional restriction, such as $Q_{\theta}(Z_{t+1} \in \cdot \mid Z_{1:t}) = Q_{\theta}(Z_{t+1} \in \cdot \mid Z_t)$, makes the observations first-order Markov. [Exercise 1](#) gives a small counterexample. This resolves the question raised by drawing one large box around the two computations: one must still specify what the box stores and what is observed.

5.1.2 The equations define a probability law

The initial law, noise laws, and maps together determine a unique law Q_{θ} on state and observation sequences. Define the transition and emission kernels

$$a_{\theta}(s, B) = \mathbb{P}(f_{\theta}(s, V_1) \in B), \quad b_{\theta}(s, D) = \mathbb{P}(g_{\theta}(s, W_1) \in D).$$

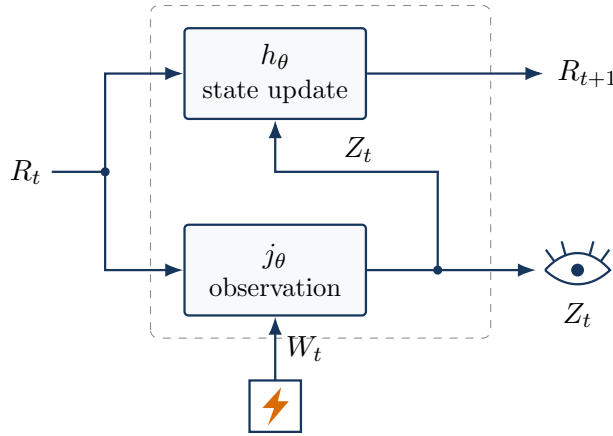


Figure 2: One step of an observation-driven model. First, R_t and the observation noise produce Z_t . Then R_t and Z_t produce R_{t+1} , which is carried to the following step. Dots split one signal into two branches. The dashed enclosure does not eliminate the state carried between steps, so Z_t need not be Markov.

Their joint law has the product form

$$Q_\theta(ds_{0:T}, dz_{1:T}) = \pi_\theta(ds_0) \prod_{t=1}^T a_\theta(s_{t-1}, ds_t) b_\theta(s_t, dz_t). \quad (5.3)$$

A density or probability-mass version uses products of transition and emission densities or masses. A general kernel need not have a density.

The correspondence is not one-to-one. Different randomized maps can implement the same kernels; different hidden representations can give the same observation law.

For a finite HMM, let $S_t \in [d]$ and $X_t \in [N]$, with

$$A_{ij} = Q_\theta(S_t = i \mid S_{t-1} = j), \quad C_{aj} = Q_\theta(X_t = a \mid S_t = j).$$

Both matrices are column-stochastic, matching column-vector probability distributions. With $\pi_j = Q_\theta(S_0 = j)$,

$$Q_\theta(s_{0:T}, x_{1:T}) = \pi_{s_0} \prod_{t=1}^T A_{s_t s_{t-1}} C_{x_t s_t}. \quad (5.4)$$

Independent uniform random variables and inverse cumulative sums of the columns implement f_θ, g_θ ; [Exercise 2](#) asks for the construction. Here X_t is the discrete observation label; later $Z_t = e_{X_t}$ is its vector encoding.

5.2 Generation after an observed prefix

Unconditional generation follows the equations: sample S_0 , then sample fresh noises at each step. Conditional generation must first account for what the observations reveal about the hidden state.

Let

$$\mu_t^- = Q_\theta(S_t \in \cdot \mid Z_{<t} = z_{<t})$$

be the *predictive state distribution*. To sample a continuation $Z_{t:T}$, first sample $S_t \sim \mu_t^-$, then sample Z_t from its emission kernel. Continue with the transition to S_{t+1} and the remaining emissions. Starting instead by applying the state update would skip the observation at time t .

After seeing z_t , Bayes' rule gives the *filtering distribution* and the next prediction:

$$\mu_t(ds) = \frac{b_\theta(z_t | s) \mu_t^-(ds)}{\int b_\theta(z_t | u) \mu_t^-(du)}, \quad \mu_{t+1}^-(B) = \int a_\theta(s, B) \mu_t(ds). \quad (5.5)$$

This notation assumes an emission density or mass and a positive denominator. Conditional laws are specified only almost surely.

For a finite HMM, $\alpha_0 = \pi$ and

$$\alpha_t^- = A\alpha_{t-1}, \quad \alpha_t = \frac{\text{diag}(C_{x_t,1}, \dots, C_{x_t,d}) \alpha_t^-}{\mathbf{1}^\top \text{diag}(C_{x_t,1}, \dots, C_{x_t,d}) \alpha_t^-}. \quad (5.6)$$

Each step costs $O(d^2)$ for dense A . This is the forward filtering recursion. Viterbi instead finds a most likely complete hidden path; smoothing computes posterior state distributions using observations on both sides of the time in question.

For continuous nonlinear models, *particle filtering* approximates the filtering distribution by weighted sampled states: propagate particles through the transition, weight them by observation likelihood, and resample when needed. It requires more than a simulator for an arbitrary observation map: the usual bootstrap filter needs evaluable likelihood weights. Stability and forgetting can prevent errors from accumulating, but dynamical stability alone is not a sufficient general theorem. Observation informativeness, mixing, dimension, and the particle count also matter; the bibliographical notes give precise sources. A Gaussian linear model permits exact filtering with means and covariances.

5.3 What can be learned from observations?

If finite hidden states were observed, transition and emission counts would give direct estimates of A, C . For linear models with observed states, least squares estimates the linear maps under standard rank conditions. Hidden states remove these directly observed targets.

We focus on a single observed trajectory $Z_{1:T}$. With independent trajectories, log likelihoods add and the recursions restart at each initial state. These are straightforward algorithmic changes; the statistical information can be substantially different.

For example, a long mixing trajectory contains repeated information about transitions but only one draw from its initial law. Even observing S_0 exactly would not consistently estimate an unrestricted π_θ from that one draw. Later observations do not supply independent initial states. Mixing and excitation help estimate the dynamics; neither manufactures new draws from π_θ . Lecture 6 returns to learning from dependent data.

5.3.1 Identifiability and the meaning of success

Relabeling hidden states leaves a finite HMM's observation law unchanged. Unreachable extra states, or suitably duplicated states with identical observable behavior, can leave it unchanged too. Thus neither the state labels nor an unrestricted state count are identifiable.

For an LDS, an invertible coordinate change $S'_t = T_0^{-1} S_t$ changes the matrices to

$$A' = T_0^{-1} A T_0, \quad C' = C T_0$$

and transforms the noise and initial laws accordingly. The observations retain exactly the same distribution. Even beyond this coordinate ambiguity, an observation law need not uniquely separate physical process noise from measurement noise.

Three useful questions are:

1. How accurately does the model predict the next observation?
2. How close are its conditional or joint observation laws, for example in KL, and its generated task statistics?
3. Under suitable minimality assumptions, how accurately are a predictive realization's matrices recovered up to a change of coordinates?

The third can help answer the first two, but the conversion needs conditioning and stability assumptions. Small matrix error alone is not a uniform long-horizon guarantee.

5.3.2 Likelihood evaluation and optimization

The observed likelihood marginalizes the hidden trajectory. For the finite HMM,

$$Q_\theta(X_{1:T}) = \sum_{s_{0:T} \in [d]^{T+1}} \pi_{s_0} \prod_{t=1}^T A_{s_t s_{t-1}} C_{X_t s_t}. \quad (5.7)$$

There are d^{T+1} terms with this initialization convention. If the initial distribution is specified directly for S_1 , there are d^T terms. The forward recursion evaluates the sum in $O(Td^2)$ arithmetic operations; normalization at each step or a log-domain calculation prevents numerical underflow. For continuous states, marginalization is an integral.

We fit θ by minimizing $-\log Q_\theta(Z_{1:T})$. Differentiating the finite-HMM forward algorithm is feasible and is implemented in modern software. Its reverse-mode differentiation has the same order of arithmetic work as likelihood evaluation, with additional storage or recomputation. It does not make the objective convex or ensure a global optimum. A large state count remains expensive.

Monte Carlo is also used for likelihoods and gradients in nonlinear state-space models. Sequential Monte Carlo exploits the time structure; sampling entire trajectories independently from the prior can produce extremely variable likelihood estimates. An unbiased likelihood estimate does not give an unbiased log likelihood after taking its log.

5.3.3 EM and optimization over hidden trajectories

Expectation-maximization (EM) uses the current parameters $\theta^{(k)}$ to compute a posterior over hidden trajectories, then maximizes the expected complete-data log likelihood:

$$\begin{aligned} q_k(ds) &= Q_{\theta^{(k)}}(ds_{0:T} \mid Z_{1:T}), \\ \theta^{(k+1)} &\in \arg \max_{\theta} \mathbb{E}_{S \sim q_k} [\log Q_\theta(S, Z_{1:T})]. \end{aligned} \quad (5.8)$$

For a finite HMM, forward-backward supplies expected transition and emission counts; these replace the observed counts. For a Gaussian LDS, the E-step uses smoothing means *and second moments*, including cross moments at adjacent times. Substituting a single estimated state path is generally a different algorithm.

Exact EM makes the observed log likelihood nondecreasing. Under regularity conditions its limit points are stationary; a local maximum of likelihood is not guaranteed in general. Efficient E- and M-steps are available for important model families, not for every nonlinear state-space model.

[Exercise 4](#) asks for the monotonicity calculation.

Another approach treats the hidden states as optimization variables:

$$\min_{\theta, s_{0:T}} -\log Q_\theta(s_{0:T}, Z_{1:T}). \quad (5.9)$$

This maximizes a joint mass or density, rather than summing or integrating over states. A prior on θ gives a joint MAP objective. Sparse trajectory optimization is central in smoothing and SLAM. It can remain nonconvex. For continuous states its density objective also depends on the chosen coordinates and reference measure.

For deterministic linear state dynamics and unit Gaussian observation noise, optimizing the initial state $s = S_1$ gives

$$\min_{A,C,s} \sum_{t=1}^T \|Z_t - CA^{t-1}s\|_2^2.$$

Unknown quantities multiply, and powers of A appear. Deterministic dynamics simplify marginalization without making this parameter fit convex.

The observation-driven model in Eq. (5.2) takes a different route: fixed R_1 and observed feedback determine every R_t , so the likelihood is a product of directly evaluable conditional observation laws. Backpropagation differentiates this deterministic computation; Section 5.8 returns to it.

5.4 Fit an observable predictor, then impose state-space structure

The proposed two-stage method first fits a flexible predictor using observed inputs and targets, then fits a compact state-space representation to that predictor. The first stage avoids hidden-state optimization. The second imposes relationships between predictions at different lags.

A finite-window model predicts from $Z_{t-p:t-1}$. If older observations have little additional predictive effect, choosing a sufficiently large p controls approximation error. This is a statement about conditional prediction, not simply about decay of the physical state.

Why fit a large, noisy model first? Many fitted coefficients can be noisy estimates of quantities determined by a few shared parameters. Enforcing those relationships can reduce estimation error. It also introduces bias when the proposed structure is wrong. Exercise 5 makes the variance reduction exact for coefficients constrained to a known low-dimensional family. The subspace algorithm below learns the family as well, so that exercise is an illustration rather than its statistical analysis.

Long memory and filtering stability can coexist. The physical state can retain information while new observations correct errors in its estimate. In an LDS, different matrices govern these two questions:

$$\text{physical or generative dynamics: } A, \quad \text{prediction-error dynamics: } F = A - LC.$$

An unstable or marginally stable A can have a stable F . Section 5.7 gives a process with permanent memory and an exact compact predictor.

5.5 Linear dynamical systems

An LDS has state $S_t \in \mathbb{R}^d$, observations $Z_t \in \mathbb{R}^m$, and equations

$$S_{t+1} = AS_t + V_{t+1}, \quad Z_{t+1} = CS_{t+1} + W_{t+1}. \quad (5.10)$$

Here $A \in \mathbb{R}^{d \times d}$, $C \in \mathbb{R}^{m \times d}$. When Gaussian noise is assumed, we say so explicitly. Linear equations alone do not specify the noise law.

Example 5.5.1 (Position and unobserved velocity). With sampling interval $\Delta > 0$, consider

$$S_t = \begin{pmatrix} P_t \\ U_t \end{pmatrix}, \quad A = \begin{pmatrix} 1 & \Delta \\ 0 & 1 \end{pmatrix}, \quad C = (1, 0), \quad V_{t+1} = \begin{pmatrix} 0 \\ \xi_{t+1} \end{pmatrix}.$$

The model takes $P_{t+1} = P_t + \Delta U_t$, then perturbs velocity. Only position is measured: $Z_t = P_t + W_t$. Differencing measurements estimates velocity, but also differences measurement noise. For independent measurement errors of variance σ_W^2 , the noise in $(Z_{t+1} - Z_t)/\Delta$ has variance $2\sigma_W^2/\Delta^2$. A state estimator can combine many measurements instead.

Both eigenvalues of A are one. State perturbations need not decay, and velocity errors accumulate in position. The process-noise covariance is singular: $\Sigma_V = \text{diag}(0, \sigma_\xi^2)$. This is appropriate for the discrete model just specified.²

Adding a small covariance εI makes a singular Gaussian transition density nonsingular and can regularize numerical calculations. It changes the model. A singular state-transition covariance does not by itself make the observation likelihood undefined: full-rank observation noise can give a nonsingular observation law. If fitted variances are allowed to collapse around interpolated data, a Gaussian likelihood can become unbounded; an appropriate variance constraint or prior addresses that issue.

We will first justify the algorithm under positive-definite process and measurement covariances and stable A . These are sufficient conditions for a short proof, not necessary conditions for Kalman filtering or subspace identification. Near-singular covariances require checking the conditional observation covariance, excitation, and the singular values used in recovery.

5.5.1 Finite HMMs also have linear vector equations

For the finite HMM, the one-hot vectors $e_{S_t} \in \mathbb{R}^d$ and $Z_t = e_{X_t} \in \mathbb{R}^N$ satisfy

$$e_{S_{t+1}} = A e_{S_t} + V_{t+1}, \quad Z_{t+1} = C e_{S_{t+1}} + W_{t+1}.$$

These have the form of Eq. (5.10), with residuals satisfying

$$\mathbb{E}[V_{t+1} \mid \mathcal{G}_t] = 0, \quad \mathbb{E}[W_{t+1} \mid \mathcal{G}_t, S_{t+1}] = 0, \quad \mathcal{G}_t = \sigma(S_{0:t}, Z_{1:t}). \quad (5.11)$$

For example, column j of A is the conditional mean of $e_{S_{t+1}}$ given $S_t = j$; subtraction defines V_{t+1} . These additive residuals generally depend on the state. They are not the independent iid random seeds in Eq. (5.1). One-hot encoding gives linear conditional means; it does not turn a finite HMM into a Gaussian LDS.

More generally, for a categorical observation,

Proposition 5.5.2 (A categorical conditional mean). *With $\mathcal{F}_t = \sigma(X_{1:t})$ and $P_t = \mathbb{E}[e_{X_{t+1}} \mid \mathcal{F}_t]$,*

$$(P_t)_a = Q(X_{t+1} = a \mid X_{1:t}), \quad Z_{t+1} = P_t + E_{t+1}, \quad \mathbb{E}[E_{t+1} \mid \mathcal{F}_t] = 0.$$

Proof. Coordinate a of $e_{X_{t+1}}$ is $\mathbb{I}\{X_{t+1} = a\}$. Take its conditional expectation, then subtract. $\square$

The residual is a *martingale difference*. With $P_t = \phi_t(X_{1:t})$, this is a vector autoregression. Squared loss against $e_{X_{t+1}}$ is the Brier loss of Lecture 3.

5.6 The regression and subspace algorithm

We now give the complete procedure before explaining why it can work. For centered real observations, choose a history length p , a block count q with $2q \leq p$, and a retained state dimension $1 \leq r \leq qm$. An intercept or a mean model is needed when centering is inappropriate. Vector norms below are Euclidean; matrix norms are induced operator norms unless marked F for Frobenius norm.

5.6.1 Fit lag coefficients

Use the observed lag vector

$$\Psi_t = \begin{pmatrix} Z_t \\ Z_{t-1} \\ \vdots \\ Z_{t-p+1} \end{pmatrix}, \quad \hat{B} = (\hat{B}_1, \dots, \hat{B}_p) \in \arg \min_B \sum_{t=p}^{T-1} \|Z_{t+1} - B\Psi_t\|_2^2. \quad (5.12)$$

Each $B_j \in \mathbb{R}^{m \times m}$ maps one observed lag to the next prediction; $B \in \mathbb{R}^{m \times mp}$. Thus

$$\hat{Z}_{t+1} = \sum_{j=1}^p \hat{B}_j Z_{t+1-j}. \quad (5.13)$$

Proposition 5.6.1 (Observable fitting is convex). *For fixed observed data, the objective in Eq. (5.12) is convex in B .*

Proof. Each prediction is linear in B ; squared Euclidean distance is convex. $\square$

This is conditional Gaussian maximum likelihood, conditional on the first p observations, for iid isotropic Gaussian residuals of fixed variance. With a specified covariance $\Omega \succ 0$, the loss is the corresponding quadratic form $u^\top \Omega^{-1} u$. For unrestricted multioutput regression its minimizer is the same ordinary least-squares coefficient matrix when the design has full rank. Other conditional noise densities give other negative-log-likelihood losses, which need not be convex.

5.6.2 Arrange, factor, and recover

Form two $qm \times qm$ block Hankel matrices:

$$\hat{\mathcal{H}}_0 = [\hat{B}_{i+j-1}]_{i,j=1}^q, \quad \hat{\mathcal{H}}_1 = [\hat{B}_{i+j}]_{i,j=1}^q. \quad (5.14)$$

Each block is $m \times m$; block rows and columns index successive lags. Equal anti-diagonals use the same fitted coefficient. This indexing explains the requirement $2q \leq p$.

Take the leading r singular values of $\hat{\mathcal{H}}_0$, assumed positive, and set

$$\hat{\mathcal{H}}_{0,r} = U_r D_r V_r^\top = \mathcal{O} \mathcal{R}, \quad \mathcal{O} = U_r D_r^{1/2}, \quad \mathcal{R} = D_r^{1/2} V_r^\top.$$

The factors have sizes $qm \times r$ and $r \times qm$. Using the Moore–Penrose pseudoinverse,

$$\bar{F} = \mathcal{O}^\dagger \hat{\mathcal{H}}_1 \mathcal{R}^\dagger, \quad \bar{C} = \mathcal{O}_{1:m,:}, \quad \bar{L} = \mathcal{R}_{:,1:m}. \quad (5.15)$$

Thus $\bar{F}$ updates an r -dimensional predictive state, $\bar{C}$ reads it out, and $\bar{L}$ inserts an observation. This is a regression-based subspace-identification construction. There are other subspace algorithms with different regressions and weighting choices.

5.6.3 Estimate innovations and generate

Run the recovered predictor on observed data:

$$\overline{M}_1 = 0, \quad \widehat{Z}_t = \overline{C}\overline{M}_t, \quad \overline{M}_{t+1} = \overline{F}\overline{M}_t + \overline{L}Z_t. \quad (5.16)$$

Compute residuals $\widehat{E}_t = Z_t - \widehat{Z}_t$. On a calibration block $\mathcal{I}$ of size n_{cal} , a zero-mean innovations fit uses

$$\widehat{\Omega} = \frac{1}{n_{\text{cal}}} \sum_{t \in \mathcal{I}} \widehat{E}_t \widehat{E}_t^\top. \quad (5.17)$$

Check residual means and temporal correlations; a persistent mean needs a mean model. Discarding an initial transient or fitting the initial predictive state can be appropriate. A covariance floor makes $\widehat{\Omega}$ positive definite when needed for a density. Using a separate block reduces reuse of fitting noise, although contiguous blocks from one trajectory are not independent.

Set $\overline{A} = \overline{F} + \overline{L}\overline{C}$ and generate

$$\widetilde{M}_1 = 0, \quad \widetilde{Z}_t = \overline{C}\widetilde{M}_t + \widetilde{E}_t, \quad \widetilde{M}_{t+1} = \overline{A}\widetilde{M}_t + \overline{L}\widetilde{E}_t, \quad \widetilde{E}_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \widehat{\Omega}). \quad (5.18)$$

This is an *innovations realization*. Its state and observation perturbations share the same $\widetilde{E}_t$; they are correlated. It is an equivalent predictive representation in the ideal case, not a recovery of independent physical process and measurement noises. Its state is an observation-based state estimate in suitable coordinates.

With exact infinite-history coefficients of a constant-gain Gaussian predictor, the residuals are indeed iid Gaussian. For finite data, truncated regression and estimated matrices, this is a modeling approximation to assess, not an automatic residual property. Moreover, finite-window population least-squares coefficients generally differ from the first p infinite-history coefficients. The omitted tail can correlate with the included lags. Increasing p reduces this bias under suitable filtering stability and design conditioning.

5.6.4 What should be checked?

Select p, q, r using held-out predictive performance and the intended use. Inspect small retained singular values, residual covariance, and both recovered matrices:

$$\rho(\overline{F}) \quad \text{for filtering,} \quad \rho(\overline{A}) \quad \text{for generation.}$$

Here ρ is spectral radius. Truncated SVD alone does not guarantee either stability condition. Predictive likelihood, generated statistics, and stability diagnostics answer different questions.

The compact model ties all lag coefficients to $\overline{C}\overline{F}^{j-1}\overline{L}$. This is the parameter sharing that the unconstrained first-stage regression omits. It can reduce variance and extend a fitted finite history to a recurrent predictor. An incorrect rank or dynamics can also worsen predictions.

5.7 A controlled example of the memory advantage

A state model can use all past measurements with fixed-size storage, whereas a fixed observation window discards information. Consider the scalar Gaussian process

$$S_t = U, \quad U \sim \mathcal{N}(0, \tau^2), \quad Z_t = U + W_t, \quad W_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2), \quad (5.19)$$

with U independent of the noises and $\tau, \sigma > 0$. Its state never forgets. The posterior mean and variance after n observations are

$$m_n = \frac{\sum_{i=1}^n Z_i}{\sigma^2/\tau^2 + n}, \quad v_n = \frac{\tau^2 \sigma^2}{\sigma^2 + n\tau^2}. \quad (5.20)$$

A recurrent predictor maintains a running sum and a count.

Proposition 5.7.1 (Full history versus a fixed window). *For $n \geq p$, the minimum mean squared prediction error for Z_{n+1} using all n observations is $\sigma^2 + v_n$. Using only the most recent p observations gives minimum error $\sigma^2 + v_p$, even among nonlinear predictors. The expected KL from the full-history predictive law to the optimal window predictive law is*

$$\frac{1}{2} \log \frac{\sigma^2 + v_p}{\sigma^2 + v_n}.$$

Proof. Gaussian conditioning gives [Eq. \(5.20\)](#). The conditional mean minimizes squared loss; the fresh measurement noise adds σ^2 . Exchangeability gives posterior variance v_p for any p observations. The expected conditional KL is $H(Z_{n+1} \mid Z_{n-p+1:n}) - H(Z_{n+1} \mid Z_{1:n})$. Use the scalar Gaussian entropy formula. $\square$

For $\tau = \sigma = 1$, $p = 10$, and $n = 1000$, the excess prediction errors above irreducible noise are $1/11$ and $1/1001$. This is a representation comparison with known model parameters. It does not claim that the subspace algorithm learns every constant-state process from one trajectory, or that a state model beats every autoregressive architecture. A recurrent state predictor itself defines an autoregressive sequence law. [Exercise 7](#) derives the formulas.

5.7.1 A finite-data comparison of the actual algorithm

A complementary experiment fits both models to the same single training trajectory. The scalar innovations generator is

$$M_{t+1} = 0.98M_t + 0.18E_t, \quad Z_t = M_t + E_t, \quad E_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

started at $M_0 = 0$ and warmed up before the retained data. Its predictor has $F = 0.8$ and coefficients $K_j = 0.18(0.8)^{j-1}$. The lag-Hankel matrix has rank one.

We compare ordinary AR(32) with its rank-one subspace fit ($q = 16, r = 1$); rank and history length are supplied, not selected using test data. Both covariance estimates use a separate 2,000-step calibration continuation. A further 10,000-step continuation evaluates conditional-mean error and Gaussian conditional KL against the exact innovations predictor. Repeating over 50 fixed random seeds measures variation between training trajectories. Results, including generation instability, appear in [Table 1](#). The reproducible script is `scripts/lecture05_synthetic.py`.

[Figure 3](#) shows held-out prediction trajectories and errors, the fitted lag coefficients, and every run's prediction risk. The trajectory is the first test segment from seed 0 at $T = 1024$, selected independently of its performance. Both fitted predictors use the same observed history. These are one-step conditional predictions, not independent free-running generated paths. The script regenerates the table, plots, and JSON results; the saved results include fitted coefficients, the plotted data, seeds, software versions, and source hashes.

At $T = 1024$ and 4096, imposing rank one substantially reduces both mean-prediction error and conditional KL. At $T = 256$, the subspace fit is worse on average and produces two unstable

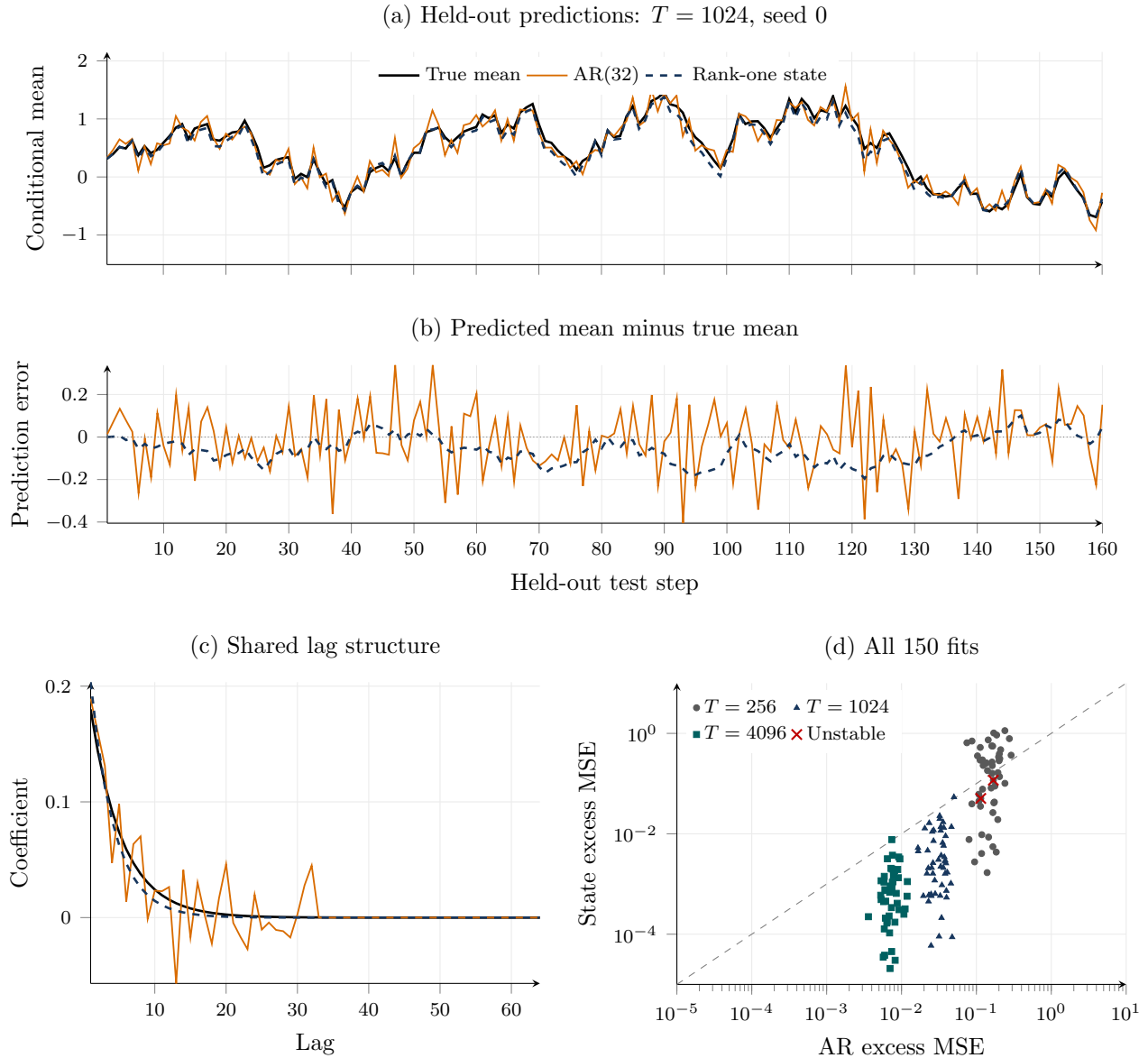


Figure 3: The experiment behind Table 1. (a,b) The first 160 test steps for the fixed seed 0 and $T = 1024$, not a run chosen for its performance. Both fitted predictors condition on the same observed history; the true conditional mean is known from the generator. The errors exclude the fresh observation noise. (c) The same fit's lag coefficients: AR(32) is zero beyond lag 32, whereas the recovered state ties all lags to $\overline{CF}^{j-1}\overline{L}$. (d) Every seed at all three training sizes. Points below the diagonal favor the state predictor. Crosses mark unstable state-space generators; they are included in the table and plot.

Training T	Model	Excess MSE	KL/step	Unstable / 50
256	AR(32)	0.1564	0.0727	0
256	Rank-one state	0.2693	0.1091	2
1024	AR(32)	0.0316	0.0158	0
1024	Rank-one state	0.0061	0.0033	0
4096	AR(32)	0.0074	0.0039	0
4096	Rank-one state	0.0011	0.0008	0

Table 1: Means over 50 trajectories. Excess MSE compares the fitted conditional mean with the true mean; irreducible variance is one. KL uses each fitted residual variance. Unstable means generative spectral radius at least one; such fits are not discarded.

generators. All fitted predictive recursions are stable in this run. The example therefore exhibits both a gain from parameter sharing and a small-sample failure of unconstrained realization.

The comparison is conditional on this low-rank data-generating model. It measures finite-data estimation, whereas [Proposition 5.7.1](#) isolates information lost by a fixed window. Neither comparison implies superiority on an arbitrary process or on language-model benchmarks.

5.8 Recurrent networks, backpropagation, and attention

A conventional recurrent predictor computes

$$R_{t+1} = h_{\theta}(R_t, u_t), \quad q_{\theta}(\cdot \mid u_{1:t}) = o_{\theta}(R_{t+1}).$$

The inputs u_t may be external measurements or previous tokens. With teacher forcing, all inputs are observed and the recurrent states are deterministic functions of them. No latent-state integral is needed to evaluate the output likelihood.

Backpropagation through time differentiates this unrolled computation. For a per-step update costing c , its work is $O(Tc)$, up to constant factors for the derivatives; straightforward storage is linear in T times the activation size. Checkpointing trades recomputation for storage. Products of state-update Jacobians can make gradients small or large. LSTM gates modify this propagation; they do not guarantee that every training problem is well-conditioned.

Linearity permits additional structure, but not a general convexity claim. For affine updates with known per-position coefficients,

$$R_{t+1} = A_t R_t + b_t, \quad (A_2, b_2) \circ (A_1, b_1) = (A_2 A_1, A_2 b_1 + b_2),$$

composition is associative. A parallel scan can reduce sequential depth. Dense matrix compositions may cost more than a sequential matrix–vector recurrence; diagonal or other structured transitions make the scan useful. Learning the matrices end to end can still be nonconvex. The coefficients must be computable without first evaluating the unknown preceding recurrent state. [Exercise 8](#) compares these operations.

Observed inputs can also be added to an LDS: $S_{t+1} = A S_t + B u_t + V_{t+1}$. Lagged inputs become additional regression features. Exogenous input assumptions and sufficient excitation matter; feedback or endogenous inputs require more care than simply adding columns.

5.8.1 An exact recurrent evaluation of linear attention

Let key and query features $k_t, q_t \in \mathbb{R}^r$, values $v_t \in \mathbb{R}^s$, and gates $\gamma_t \in \mathbb{R}$ be computed from a layer's inputs. The matrix state

$$R_0 = 0, \quad R_t = \gamma_t R_{t-1} + v_t k_t^\top, \quad y_t = R_t q_t \quad (5.21)$$

has sr coordinates.

Proposition 5.8.1 (Recurrent and pairwise evaluations). *The same output is*

$$y_t = \sum_{u=1}^t v_u (k_u^\top q_t) \prod_{j=u+1}^t \gamma_j.$$

Proof. Unroll the affine recursion for R_t and multiply by q_t . The empty product equals one. $\square$

For $\gamma_t = 1$, this is unnormalized causal linear attention. For nonnegative kernel weights, a normalized version maintains $d_t = d_{t-1} + k_t$ and uses $R_t q_t / (d_t^\top q_t)$, when the denominator is positive. The state and recurrent work per token are $O(sr)$, in addition to feature computation. Directly evaluating all pairwise interactions has quadratic dependence on sequence length.

This identity concerns a finite-feature attention map. Unrestricted softmax attention need not admit such an exact fixed-size summary.³

5.8.2 What this says about language models

One can train recurrent-compatible layers from the start, or convert a trained transformer using distillation and further training. Mamba-2 develops structured state-space/attention computations with efficient blocked implementations. MOHAWK matches mixing matrices, intermediate representations, and predictions in stages. Mamba-in-Llama reuses attention projections and trains hybrid models; retained attention layers still have growing caches.

Hazan and collaborators' spectral filtering takes another route: fit combinations of fixed temporal filters whose span approximates a specified family of dynamical predictors. It can avoid identifying a physical transition matrix. Spectral state-space sequence layers extend this idea, and observation spectral filtering addresses prediction from outputs. These methods should be distinguished from recovering (F, C, L) by the Hankel procedure.

A small recurrent state offers economical sequential inference. It must also support the information retrieval, optimization, numerical stability, and accelerator utilization needed by the task. The classical statistical guarantees below do not establish a winner between transformer and state-space language models. The cited papers demonstrate particular constructions and experiments; broad architectural dominance remains an empirical question. The possibility of stronger state-space models is a research motivation.

5.9 Why the algorithm can work*

The following results justify the construction under explicit assumptions. They may be read after the algorithm and classroom discussion.

5.9.1 Gaussian prediction and truncation

For a Gaussian LDS, the conditional distribution can be computed by a recursive predictor. Its state is an estimate obtained from the observations. This gives an observable autoregression after unrolling. The tail of its history coefficients controls the truncation error.

The observations in this subsection are real vectors. For their laws, log loss uses a probability density, and KL is

$$D(P, Q) = \int p(z) \log \frac{p(z)}{q(z)} dz,$$

where p, q are the densities of P, Q . The probability and KL chain rules apply to densities in the same way.

Theorem 5.9.1 (A constant-gain Gaussian predictor). *Initialize the system at time 1 for this theorem. Consider $S_{t+1} = AS_t + V_{t+1}$, $Z_t = CS_t + W_t$ for $t \geq 1$, with spectral radius $\rho(A) < 1$, independent centered Gaussian process and observation noises with covariances $\Sigma_V \succ 0$ and $\Sigma_W \succ 0$. There are matrices $\Pi \succ 0$ and L such that, when $S_1 \sim \mathcal{N}(0, \Pi)$ independently of the noises, the recursion*

$$M_1 = 0, \quad M_{t+1} = FM_t + LZ_t, \quad F = A - LC \quad (5.22)$$

satisfies

$$M_t = \mathbb{E}[S_t \mid Z_{<t}], \quad Z_t \mid Z_{<t} \sim \mathcal{N}(CM_t, \Omega), \quad \Omega = C\Pi C^\top + \Sigma_W. \quad (5.23)$$

Moreover, $\rho(F) < 1$ and the observations have uniformly bounded second moments.

The recursion is the *Kalman predictor*. The covariance Π is chosen to make its gain constant; [Appendix 5.A.1](#) constructs it and proves the theorem. For an arbitrary Gaussian initialization, the same conditional calculation gives time-dependent gains. Positive noise covariance means positive variance in every nonzero direction. A stable A permits partially observed states and arbitrary C in this result.

Define the *innovations* $E_t = Z_t - CM_t$. Their conditional law is always $\mathcal{N}(0, \Omega)$, so they are independent across time and independent of the previous observations. Consequently, the same observation law can be generated by

$$Z_t = CM_t + E_t, \quad M_{t+1} = (F + LC)M_t + LE_t = AM_t + LE_t. \quad (5.24)$$

The conditional predictor and its innovations law together specify generation.

Unrolling [Eq. \(5.22\)](#) gives

$$CM_t = \sum_{j=1}^{t-1} K_j Z_{t-j}, \quad K_j = CF^{j-1}L. \quad (5.25)$$

Because F is stable, these coefficients decay geometrically. The following proposition translates their tail into a sequence-law guarantee.

Proposition 5.9.2 (Truncation controls prediction and sequence KL). *In the setting above, suppose $\|K_j\| \leq G\gamma^{j-1}$ for $j \geq 1$, with $0 < \gamma < 1$, and $\sup_t (\mathbb{E}[\|Z_t\|_2^2])^{1/2} \leq B_Z$. Define*

$$\widehat{Z}_t^{(p)} = \sum_{j=1}^{\min(p, t-1)} K_j Z_{t-j}.$$

Then

$$(\mathbb{E}[\|CM_t - \hat{Z}_t^{(p)}\|_2^2])^{1/2} \leq a_p := \frac{GB_Z\gamma^p}{1-\gamma}. \quad (5.26)$$

Let $P_{1:T}$ be the true observation law and let $Q_{1:T}^{(p)}$ use conditional law $\mathcal{N}(\hat{Z}_t^{(p)}, \Omega)$. Then

$$D(P_{1:T}, Q_{1:T}^{(p)}) \leq \frac{Ta_p^2}{2\lambda_{\min}(\Omega)}. \quad (5.27)$$

Proof. The triangle inequality in L^2 bounds the omitted tail by $B_Z \sum_{j>p} G\gamma^{j-1} = a_p$. For Gaussians with the same covariance,

$$D(\mathcal{N}(u, \Omega), \mathcal{N}(v, \Omega)) = \frac{1}{2}(u - v)^\top \Omega^{-1}(u - v).$$

Apply this at each observed prefix and use the KL chain rule. $\square$

This explains why an observable autoregression can approximate a noisy state-space model well: the lost remote history has an explicit cost. These bounds use the true coefficients. Regression contributes coefficient-estimation error in addition to this truncation error. Squared loss is convenient for the convex fit and the Gaussian calculation; the state-elimination identity itself is independent of the fitting loss.

5.9.2 Exact recovery from ideal coefficients

The coefficients in [Eq. \(5.25\)](#) come from a small recurrent computation. Their matrix structure reveals that computation. Suppose $K_j = C_\star F_\star^{j-1} L_\star$ has minimal state dimension $d_\star$. Choose a block count q for which

$$\mathcal{O}_\star = \begin{pmatrix} C_\star \\ C_\star F_\star \\ \vdots \\ C_\star F_\star^{q-1} \end{pmatrix}, \quad \mathcal{R}_\star = (L_\star, F_\star L_\star, \dots, F_\star^{q-1} L_\star)$$

both have rank $d_\star$. The first maps a state to successive readouts; the second collects how successive observations enter the state. Form the $qm \times qm$ block matrices

$$\mathcal{H}_0 = [K_{i+j-1}]_{i,j=1}^q = \mathcal{O}_\star \mathcal{R}_\star, \quad \mathcal{H}_1 = [K_{i+j}]_{i,j=1}^q = \mathcal{O}_\star F_\star \mathcal{R}_\star. \quad (5.28)$$

Each block is $m \times m$; block rows and columns index successive lags. A matrix constant along block anti-diagonals is a *block Hankel matrix*. Here $\text{rank}(\mathcal{H}_0) = d_\star$.

Take its rank- $d_\star$ singular value decomposition and factor it:

$$\mathcal{H}_0 = UDV^\top = \mathcal{O}\mathcal{R}, \quad \mathcal{O} = UD^{1/2}, \quad \mathcal{R} = D^{1/2}V^\top.$$

Apply [Eq. \(5.15\)](#) with the exact matrix $\mathcal{H}_1$ and these factors. Then the matrices satisfy the following recovery property.

Proposition 5.9.3 (Recovery of the prediction map). *The recovered matrices satisfy*

$$\overline{C} \overline{F}^{j-1} \overline{L} = K_j \quad \text{for every } j \geq 1.$$

Thus [Eq. \(5.16\)](#) computes the same predictor on every observed history.

The proof in [Appendix 5.A.2](#) shows that the recovered states differ from minimal predictive coordinates by an invertible change of basis. Their role is to implement prediction; recovering the original physical coordinates is optional.

5.9.3 What finite-data guarantees must control

Exact factorization explains the algorithm, but says nothing by itself about how many observations are needed. Four errors must be controlled: finite-history approximation, estimation of the regression coefficients, matrix realization, and estimation of the innovations law.

The regression error has a useful exact decomposition. Write

$$Z_{t+1} = B^{[p]}\Psi_t + r_t + E_{t+1}, \quad B^{[p]} = (K_1, \dots, K_p),$$

where r_t is the omitted predictor tail and E_{t+1} is the innovation. If $\sum_t \Psi_t \Psi_t^\top$ is invertible, ordinary least squares gives

$$\hat{B} - B^{[p]} = \left(\sum_t E_{t+1} \Psi_t^\top + \sum_t r_t \Psi_t^\top \right) \left(\sum_t \Psi_t \Psi_t^\top \right)^{-1}. \quad (5.29)$$

The first numerator is a martingale sum. The second is truncation error correlated with the regressors. The inverse Gram matrix expresses *excitation*: insufficient variation in some direction makes that direction hard to estimate. Dependence between overlapping windows must be handled in the probability bounds.

If every coefficient used in a q -block Hankel matrix has operator error at most δ , its matrix error is at most $q\delta$. Indeed, each output block has norm at most $\delta \sum_j \|x_j\|$, and Cauchy–Schwarz bounds the full norm. If the true Hankel matrix has rank r , the best rank- r approximation to its estimate obeys

$$\|\hat{\mathcal{H}}_{0,r} - \mathcal{H}_0\| \leq 2\|\hat{\mathcal{H}}_0 - \mathcal{H}_0\|. \quad (5.30)$$

This follows because the discarded singular value is no larger than the perturbation norm. Converting this to matrices requires controlling $\sigma_r(\mathcal{H}_0)$ and the pseudoinverses. Low-rank approximation does not by itself prove improved prediction risk.

There are finite-sample theorems for related stochastic subspace algorithms, including single-trajectory recovery up to similarity at order $T^{-1/2}$ with logarithmic and system-dependent factors. Some allow $\rho(A) \leq 1$, under observability, controllability by the process noise, positive measurement covariance, known order, and a steady-state filtering initialization. Their future-on-past block regression is different from the one-step AR fit written here. The bibliographical notes give the exact source and theorem locations. We do not transfer its rate to this implementation without analyzing Eq. (5.29) and its realization step.

5.9.4 From matrix errors to prediction and sequence laws

The following calculation makes a useful part of the conversion explicit. Compare a true and fitted predictor on the *same observed history*, in aligned state coordinates:

$$M_{t+1} = FM_t + LZ_t, \quad \bar{M}_{t+1} = \bar{F}\bar{M}_t + \bar{L}Z_t, \quad M_1 = \bar{M}_1 = 0.$$

Suppose $\|\bar{F}\| \leq \rho < 1$, the true states and observations have RMS norms bounded by B_M, B_Z , and $\|\bar{F} - F\| \leq \delta_F$, $\|\bar{L} - L\| \leq \delta_L$, $\|\bar{C} - C\| \leq \delta_C$. Then the state discrepancy e_t obeys

$$e_{t+1} \leq \rho e_t + \delta_F B_M + \delta_L B_Z.$$

Consequently

$$\sup_t (\mathbb{E}[\|\bar{C}\bar{M}_t - CM_t\|^2])^{1/2} \leq \varepsilon_\mu := \frac{\|\bar{C}\|(\delta_F B_M + \delta_L B_Z)}{1 - \rho} + \delta_C B_M. \quad (5.31)$$

A weighted norm may supply contraction when a stable matrix does not contract the ordinary Euclidean norm. Alignment and norm conditioning then affect the constants.

For Gaussian conditional laws, mean accuracy and covariance accuracy together give sequence KL.

Proposition 5.9.4 (Conditional errors control Gaussian sequence KL). *Let P and Q have conditional laws $\mathcal{N}(\mu_t, \Omega)$ and $\mathcal{N}(\hat{\mu}_t, \hat{\Omega})$, evaluated at the same prefix, with positive-definite constant covariances. Then*

$$D(P_{1:T}, Q_{1:T}) = \frac{1}{2} \sum_{t=1}^T \mathbb{E}_P[(\mu_t - \hat{\mu}_t)^\top \hat{\Omega}^{-1}(\mu_t - \hat{\mu}_t)] + \frac{T}{2} \left[\text{tr}(\hat{\Omega}^{-1}\Omega) - m + \log \frac{\det \hat{\Omega}}{\det \Omega} \right].$$

If $\mathbb{E}_P[\|\mu_t - \hat{\mu}_t\|^2] \leq \varepsilon_\mu^2$ and all eigenvalues of $\hat{\Omega}^{-1/2}\Omega\hat{\Omega}^{-1/2}$ lie in $[1 - \eta, 1 + \eta]$, $0 \leq \eta < 1$, then

$$\frac{1}{T} D(P_{1:T}, Q_{1:T}) \leq \frac{\varepsilon_\mu^2}{2\lambda_{\min}(\hat{\Omega})} + \frac{m\eta^2}{4(1 - \eta)^2}. \quad (5.32)$$

Proof. Apply the sequence KL chain rule and the Gaussian KL formula. For $\lambda \in [1 - \eta, 1 + \eta]$, Taylor's theorem gives $\lambda - 1 - \log \lambda \leq (\lambda - 1)^2/[2(1 - \eta)^2]$. Sum this bound over the covariance eigenvalues. $\square$

This explains which small-noise issue matters for KL: mean errors are measured in units of conditional observation variance. A nearly zero $\lambda_{\min}(\hat{\Omega})$ can make a small Euclidean error costly. A singular physical Σ_V alone does not force this problem.

For any sequence cost with range length $C_{\max}$, Pinsker gives

$$|\mathbb{E}_P[c] - \mathbb{E}_Q[c]| \leq C_{\max} \sqrt{D(P_{1:T}, Q_{1:T})/2}.$$

Uniformly small per-step KL need not give a small discrepancy for arbitrarily long sequences. To control long-run generated statistics, use stability of the *generative* dynamics and an appropriate coupling. [Appendix 5.A.3](#) gives that comparison. A prediction theorem about $\bar{F}$ does not by itself certify the needed condition on $\bar{A} = \bar{F} + \bar{L}\bar{C}$.

5.9.5 Stable autoregression can also control long-run error

Stability can be imposed directly on the observable regression. Here is one sufficient condition that preserves convex fitting.

Proposition 5.9.5 (A stable autoregressive fit). *Let $Z_{t+1} = P_t + E_{t+1}$, where P_t is its conditional mean and the E_t are iid additive innovations independent of the past. Fix fitted coefficient matrices satisfying*

$$\sum_{j=1}^p \|\hat{B}_j\| \leq \beta < 1, \quad \sup_{t \geq p} \left(\mathbb{E} \left[\left\| P_t - \sum_{j=1}^p \hat{B}_j Z_{t+1-j} \right\|_2^2 \right] \right)^{1/2} \leq \varepsilon.$$

Start a fitted generator from the same first p observations and use the same innovations in a coupling:

$$\tilde{Z}_{t+1} = \sum_{j=1}^p \hat{B}_j \tilde{Z}_{t+1-j} + E_{t+1}.$$

Then

$$\sup_t (\mathbb{E}[\|Z_t - \tilde{Z}_t\|_2^2])^{1/2} \leq \frac{\varepsilon}{1 - \beta}. \quad (5.33)$$

Proof. Let d_t denote the RMS discrepancy. Inserting the fitted predictor evaluated on the true history and using the L^2 triangle inequality gives $d_{t+1} \leq \varepsilon + \sum_j \|\hat{B}_j\| d_{t+1-j}$. Starting from $d_1 = \dots = d_p = 0$, induction gives the result. $\square$

The constraint on the sum of operator norms is convex. An L_c -Lipschitz statistic of one observation, or its time average, then has expectation discrepancy at most $L_c \varepsilon / (1 - \beta)$. [Appendix 5.A.3](#) gives the corresponding state-space comparison and a second-moment bound. Thus both state-space modeling and autoregression can use stability. The choice concerns how useful memory is represented, how much is truncated, and what constraints the fitting method preserves.

5.9.6 A noiseless comparison: Cayley–Hamilton

In a noiseless LDS, we can eliminate the hidden state to obtain an exact recurrence involving only observations. The key linear-algebra result is the following.

Theorem 5.9.6 (Cayley–Hamilton). *Let $A \in \mathbb{R}^{d \times d}$ have characteristic polynomial*

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^d + a_{d-1}\lambda^{d-1} + \dots + a_1\lambda + a_0.$$

Then

$$\chi_A(A) = A^d + a_{d-1}A^{d-1} + \dots + a_1A + a_0I = 0. \quad (5.34)$$

The proof by the adjugate identity is in [Appendix 5.A.4](#).

Corollary 5.9.7 (Eliminating a noiseless hidden state). *If $S_{t+1} = AS_t$ and $Z_t = CS_t$, then*

$$Z_{t+d} + a_{d-1}Z_{t+d-1} + \dots + a_0Z_t = 0. \quad (5.35)$$

Proof. The left side is $C\chi_A(A)S_t = 0$. $\square$

Even dynamics that retain information indefinitely have this recurrence. No forgetting assumption was used. With noise, the same elimination produces additional, generally correlated noise terms. To fit a predictor by regression, we need its conditional mean given the *observations*, as in [Section 5.9.1](#). Thus state elimination alone does not justify the noisy subspace method.

5.10 Take-home messages

1. A useful state-space model restricts what is remembered and how it evolves. The observation process need not be Markov.
2. Conditional generation first infers a distribution over the current state. Filtering, smoothing, and most likely path decoding answer different questions.
3. Hidden-state likelihood can be evaluated efficiently in important families, while its optimization remains difficult. EM uses posterior averages; joint state optimization uses a different objective.
4. Observable regression supplies a flexible predictor. Hankel factorization imposes shared dynamics and can recover a compact innovations realization. Finite-data success depends on truncation, excitation, rank conditioning, and residual modeling.
5. Memory in the physical state and forgetting of filtering errors are compatible. Prediction stability and generation stability must be checked separately.
6. Deterministic recurrent states permit backpropagation through observed prefixes. Structured affine updates can also support parallel training and recurrent evaluation of some attention maps.

Endnotes

1. **General spaces.** Standard Borel spaces suffice for the conditional distributions and randomized kernel representations used here. An arbitrary measurable space need not have these properties.
2. **When does acceleration also perturb position?** If a random acceleration acts throughout a sampling interval, integrating it generally gives correlated position and velocity perturbations. The zero position-noise convention corresponds to motion at the old velocity followed by a random velocity change at the interval boundary.
3. **A rank obstruction.** Choose scalar queries $q_i = i - 1$ and keys $k_j = \log a_j$ for distinct positive a_j . Then $[\exp(q_i k_j)]_{i,j=1}^n = [a_j^{i-1}]_{i,j=1}^n$ is an invertible Vandermonde matrix. Row normalization preserves rank. Putting these keys before these queries gives a past-to-future block of causal attention with rank n . A fixed r -feature bilinear kernel has rank at most r , so it cannot represent all these blocks for $n > r$. This is a restriction on finite-feature representations, not a theorem about every conceivable recurrent computation.

Bibliographical notes

Filtering, smoothing, and likelihood fitting. Doucet and Johansen [2009] give an accessible particle-filtering and smoothing tutorial. Douc et al. [2014] establishes time-uniform control of asymptotic particle-filter variance and error under mixing and observation assumptions, including noncompact state spaces. These results concern a specified filtering problem and particle approximation; they do not assert that every stable dynamical system is easy to filter in high dimension.

Dynamax implements EM and gradient-based marginal-likelihood fitting [Dynamax, 2026]. Thus differentiating a forward recursion is an established computational option; no general empirical claim of its failure is intended here. Wu [1983] supplies convergence conditions for EM and explains why monotonicity alone does not establish convergence to a local optimum. Dellaert and Kaess [2017] develops sparse trajectory MAP estimation for robotics. It concerns optimization over states and, in SLAM, map variables, rather than replacing marginal likelihood by an equivalent objective.

Gaussian prediction and realization. Mikusheva [2009] derives Gaussian state prediction. The sufficient-condition proof in Appendix 5.A.1 is included in full. Ho and Kalman [1966] gives the classical realization construction from input-output coefficient matrices; Section 5.9.2 applies that algebra to the predictor’s coefficients.

For finite stochastic data, Tsiamis and Pappas [2019], Theorem 1 and Theorem 4 with Remark 4, analyzes future-on-past block regression and balanced realization. Assumptions 1–3 specify known order, observability, controllability by process noise, positive measurement covariance, steady-state filtering initialization, and sufficient block lengths. Together with sample-size and conditioning requirements, the bounds give matrix error of order $T^{-1/2}$ up to logarithmic factors and similarity. The allowed $\rho(A) \leq 1$ includes marginal stability. This is a closely related algorithm, not a finite-sample theorem for every one-step AR/SVD variant.

Tsiamis and Pappas [2020] instead obtains high-probability polylogarithmic cumulative squared-loss regret relative to the Kalman predictor using online regression, including non-explosive systems under the paper’s conditions. This directly addresses prediction performance; it need not identify a compact generative realization. The Gaussian KL and coupling calculations here explain additional steps needed when the evaluation criterion is a sequence law or a generated statistic.

Synthetic comparisons and spectral prediction. Synthetic dynamical systems are a standard controlled setting for comparing sequence models. Agarwal et al. [2024], Section 4.1 and Appendix E.1, tests a system with diagonal eigenvalues ± 0.9999 and compares spectral units with learned linear recurrence. These eigenvalues are strictly inside the unit circle, although the appendix calls the system marginally stable. That experiment is not a proof that subspace identification beats optimal finite-window prediction. Our constant-state calculation and the reproducible AR/SVD experiment make those comparisons separately.

Hazan et al. [2017] fits a predictor in a fixed spectral feature space and proves guarantees against a specified LDS class. The learner need not recover the comparator’s state coordinates. Agarwal et al. [2024] develops spectral sequence layers and an autoregressive extension. Dogariu et al. [2025] extends observable-history prediction using spectral filtering and linear observers in a nonlinear-dynamics setting; the bounds require their structural and conditioning assumptions. These are additional ways to exploit dynamical structure, not statistical guarantees for arbitrary language processes.

Recurrent computation and language models. Pascanu et al. [2013] analyzes vanishing and exploding gradients in recurrent-network training. Katharopoulos et al. [2020] derives recurrent causal linear attention. Dao and Gu [2024], Sections 2 and 5, develops structured state-space duality and efficient matrix-based algorithms; the finite-feature calculation in Section 5.8.1 is an elementary example.

Bick et al. [2024] introduces MOHAWK’s staged transformer-to-SSM distillation. Wang et al. [2024] converts pretrained attention layers and trains hybrid recurrent/attention models. Their measured results concern the models, tasks, and compute settings in those papers. Classical LDS consistency results do not imply that these neural distillation procedures recover an exact dynamical system.

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Glossary

State-space model; latent state

A model of observations through an evolving, possibly unobserved state. State coordinates specify the information retained over time.

One-hot encoding; martingale difference

$Z_t = e_{X_t}$ turns token probabilities into vector conditional means. Subtracting a conditional mean produces a residual with conditional expectation zero given the past.

Linear dynamical system; linear autoregression

An LDS has linear state and observation equations. A linear autoregression predicts from a linear combination of observed lags.

System identification

Estimating dynamical and observation equations from observed data.

Innovation; Kalman predictor

An innovation is the observation minus its conditional-mean prediction. For the Gaussian LDS considered here, the Kalman predictor computes that mean recursively, and innovations are independent Gaussian vectors.

Truncation error

The predictive effect of history omitted by a finite-lag approximation. It is controlled by the tail of the history coefficients.

Hankel factorization

Factoring a matrix of lag coefficients to recover a finite-dimensional recurrent predictor. Its rank expresses the minimal predictive state dimension under the stated rank assumptions.

Filtering; prediction; smoothing

Conditioning a current state on observations through the current time, conditioning a future state on earlier observations, and conditioning a state on observations before and after it.

Observation-driven state

A recurrent state computed deterministically from an observed prefix.

EM

Alternating posterior inference and expected complete-data likelihood maximization.

Excitation; realization conditioning

Variation in observed regressors and separation of retained singular values from zero; both affect recovery from noisy data.

Backpropagation through time

Differentiation of the unrolled recurrent computation.

5.11 Exercises

The exercises follow the modeling, estimation, and computational questions. The noiseless recurrence in [Exercise 6\(a\)](#) is optional.

General instructions. Justify every answer, including yes/no answers, even when the question does not explicitly ask for a justification.

1. **Observed feedback does not imply Markov observations.** Let W_t be iid Bernoulli(1/4), put $Z_{-1} = Z_0 = 0$, and set $Z_t = Z_{t-2} \oplus W_t$, where $\oplus$ is addition modulo two.
 - (a) Express the model as [Eq. \(5.2\)](#) with $R_t = (Z_{t-2}, Z_{t-1})$. Show that (R_t, Z_t) is Markov.
 - (b) Compare $Q(Z_3 = 1 \mid Z_1 = 0, Z_2 = 0)$ and $Q(Z_3 = 1 \mid Z_1 = 1, Z_2 = 0)$. Prove that both conditioning events have positive probability and that (Z_t) is not first-order Markov.
2. **From noise-driven maps to an HMM law.** Use [Eqs. \(5.1\)](#) and [\(5.3\)](#) and the finite HMM matrices A, C defined in [Eq. \(5.4\)](#).
 - (a) Prove the joint-law product formula by successive conditioning. Write the marginal law of $S_{1:T}, Z_{1:T}$ after integrating out S_0 .
 - (b) For independent uniform $[0, 1]$ noises, give explicit inverse-cumulative-sum maps implementing A and C , including a convention at zero-probability endpoints.
 - (c) Construct two different noise maps with the same transition kernel. Explain why a specified model gives a unique law but the law does not uniquely specify its noise maps.
 - (d) Suppose the hidden states are observed. Derive the column-normalized transition and emission count estimates. Which columns remain undetermined if some states are never visited?
3. **Latent likelihood and the forward algorithm.** Use the hidden Markov model in [Eq. \(5.7\)](#), with d hidden states, transition matrix A , emission matrix C , and initial law π .
 - (a) Derive the likelihood by summing the joint law of hidden states and observations. How many hidden trajectories appear?
 - (b) Starting from $\Lambda_0 = \pi$, derive the recursion

$$\Lambda_t = \text{diag}(C_{X_{t,1}}, \dots, C_{X_{t,d}}) A \Lambda_{t-1}, \quad Q_\theta(X_{1:T}) = \mathbf{1}^\top \Lambda_T.$$

Prove that $\Lambda_t(j)$ is the joint mass the model assigns to the observed prefix and hidden state j at time t . Explain the $O(Td^2)$ arithmetic cost.

- (c) If $S_{0:T}$ were observed, write the transition and emission counts that appear in the complete-data log likelihood. What replaces those counts when the hidden trajectory is averaged under its posterior given $X_{1:T}$?
 - (d) For a continuous state model with initial density π_θ , transition density $a_\theta(s' \mid s)$, and emission density $b_\theta(x \mid s)$, write the corresponding trajectory integral.
4. **EM averages hidden paths.** In a finite model with positive joint masses, fix observed data z , write $\ell(\theta) = \log Q_\theta(z)$, and let $q(s) = Q_{\theta_0}(s \mid z)$.
 - (a) Prove

$$\ell(\theta) - \ell(\theta_0) = \sum_s q(s) \log \frac{Q_\theta(s, z)}{Q_{\theta_0}(s, z)} + D(q, Q_\theta(\cdot \mid z)).$$

Deduce monotonicity of exact EM and of an M-step that merely increases the expected complete-data log likelihood.

- (b) Why does this identity not guarantee convergence to a local maximum? Explain why replacing q by a point mass at a most likely path generally changes the update and its guarantee.
 - (c) For a Gaussian LDS, expand $\mathbb{E}[\|S_{t+1} - AS_t\|^2 \mid Z_{1:T}]$. Identify the second moments that a state-mean substitution omits.
5. **Denoising coefficients by parameter sharing.** Let $U \in \mathbb{R}^{p \times r}$ be known with $U^\top U = I_r$. Suppose a preliminary estimate is $\hat{b} = b + \xi$, where $\xi \sim \mathcal{N}(0, \sigma^2 I_p/n)$. Set $\tilde{b} = UU^\top \hat{b}$.
- (a) If b lies in the column space of U , compute the expected squared coefficient errors of $\hat{b}$ and $\tilde{b}$.
 - (b) Let an independent test feature Ψ have mean zero and $\mathbb{E}[\Psi\Psi^\top] = I_p$. Compute the corresponding prediction errors for predicting $b^\top \Psi$.
 - (c) Realize a one-dimensional example using $b_j = \theta \lambda^{j-1}$, $j = 1, \dots, p$, with known $|\lambda| < 1$. Explain the relationship to a scalar recurrent predictor.
 - (d) For arbitrary b , derive the bias-variance formula for $\tilde{b}$. Exactly when is its expected coefficient error smaller than that of $\hat{b}$?

The independent isotropic coefficient errors are a simplifying model, not an assertion about lagged time-series least squares.

6. **Observable recurrence and predictive state.*** Use $M_{t+1} = FM_t + LZ_t$, $\hat{Z}_t = CM_t$, and $K_j = CF^{j-1}L$ from Eqs. (5.22) and (5.25).
- (a) Let $S_{t+1} = AS_t$, where $A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, and let $Z_t = c^\top S_t$. Use Cayley-Hamilton to derive an order-two recurrence for Z_t . Show that distinct initial states remain at the same Euclidean distance under these dynamics.
 - (b) Unroll Eq. (5.22) from a nonzero fixed M_1 . Which initialization term must be added to Eq. (5.25)?
 - (c) Verify the two block factorizations in Eq. (5.28), including the index shifts.
 - (d) Show directly that replacing (F, C, L) by $(T_0^{-1}FT_0, CT_0, T_0^{-1}L)$, with T_0 invertible, preserves every history coefficient.
 - (e) For scalars $F = 1/2, C = 1, L = 3/4$, compare predictor-memory decay with the dynamics used for generation. Which matrix is relevant to each question?
7. **Permanent state and finite-window prediction.** Use the process in Eq. (5.19).
- (a) Complete the square in the posterior density of U given $Z_{1:n}$ to derive Eq. (5.20). Give a recursion that stores only a running sum and a count.
 - (b) Derive the minimum prediction risks using all n observations and only the last $p \leq n$. Prove that the latter bound applies to nonlinear predictors as well as linear AR predictors.
 - (c) Derive the expected conditional KL in Proposition 5.7.1 directly from the Gaussian KL formula. You may use conditional-expectation orthogonality.
 - (d) Explain why recovering U accurately along one infinite trajectory does not consistently estimate an unknown prior variance τ^2 . What additional data would supply repeated draws of U ?

8. **Backpropagation, affine scans, and recurrent attention.** Consider $R_{t+1} = h_\theta(R_t, u_t)$ with observed inputs.

- (a) Express $\partial R_T / \partial R_s$ as a product of state-update Jacobians. For a scalar linear update $R_{t+1} = aR_t$, compare $0 < |a| < 1$ and $|a| > 1$.
- (b) Verify associativity of the affine composition in [Section 5.8](#). Compare a dense d -dimensional sequential recurrence and a parallel scan in work, depth, and activation storage. What improves if the transition matrices are diagonal?
- (c) Show that linearity in the state does not make learning the transition parameter convex: use $R_0 = 1$, $R_2 = a^2$, and loss $(R_2 - 1)^2$.
- (d) Derive [Proposition 5.8.1](#). For $\gamma_t = 1$, derive the extra denominator state needed for normalized linear attention. State when the denominator is valid.

5.A Deferred proofs and stability comparisons

5.A.1 The constant-gain Gaussian predictor

Theorem 5.9.1 (A constant-gain Gaussian predictor). *Initialize the system at time 1 for this theorem. Consider $S_{t+1} = AS_t + V_{t+1}$, $Z_t = CS_t + W_t$ for $t \geq 1$, with spectral radius $\rho(A) < 1$, independent centered Gaussian process and observation noises with covariances $\Sigma_V \succ 0$ and $\Sigma_W \succ 0$. There are matrices $\Pi \succ 0$ and L such that, when $S_1 \sim \mathcal{N}(0, \Pi)$ independently of the noises, the recursion*

$$M_1 = 0, \quad M_{t+1} = FM_t + LZ_t, \quad F = A - LC \quad (5.22)$$

satisfies

$$M_t = \mathbb{E}[S_t \mid Z_{<t}], \quad Z_t \mid Z_{<t} \sim \mathcal{N}(CM_t, \Omega), \quad \Omega = C\Pi C^\top + \Sigma_W. \quad (5.23)$$

Moreover, $\rho(F) < 1$ and the observations have uniformly bounded second moments.

Proof. We construct the covariance used in [Theorem 5.9.1](#). If $S_t \mid Z_{<t} \sim \mathcal{N}(M_t, \Pi_t)$, Gaussian conditioning and the state equation give

$$\begin{aligned} \Omega_t &= C\Pi_t C^\top + \Sigma_W, & L_t &= A\Pi_t C^\top \Omega_t^{-1}, \\ M_{t+1} &= AM_t + L_t(Z_t - CM_t), \\ \Pi_{t+1} &= A\Pi_t A^\top + \Sigma_V - A\Pi_t C^\top \Omega_t^{-1} C\Pi_t A^\top. \end{aligned} \quad (5.36)$$

The covariance update is a deterministic map $\mathcal{T}(\Pi_t)$. To show that it has a fixed point under the stated conditions, use the identity

$$(A - GC)\Pi(A - GC)^\top + \Sigma_V + G\Sigma_W G^\top = \mathcal{T}(\Pi) + (G - L)(C\Pi C^\top + \Sigma_W)(G - L)^\top, \quad (5.37)$$

where $L = A\Pi C^\top (C\Pi C^\top + \Sigma_W)^{-1}$. For any vector, its quadratic form on the left is minimized over G by $G = L$. It follows that $\mathcal{T}$ is monotone in the positive-semidefinite order.

Since $\rho(A) < 1$, the finite matrix

$$B = \sum_{j=0}^{\infty} A^j \Sigma_V (A^\top)^j$$

satisfies $B = ABA^\top + \Sigma_V$. Starting from $\Pi_0 = 0$, iterate $\Pi_{n+1} = \mathcal{T}(\Pi_n)$. Monotonicity gives an increasing sequence, and $\mathcal{T}(B) \preceq ABA^\top + \Sigma_V = B$ bounds it above. It therefore converges to a fixed point Π . Continuity follows from $\Sigma_W \succ 0$, and $\Pi \succeq \Sigma_V \succ 0$.

Set $L = A\Pi C^\top (C\Pi C^\top + \Sigma_W)^{-1}$ and $F = A - LC$. The fixed-point identity becomes

$$\Pi = F\Pi F^\top + \Sigma_V + L\Sigma_W L^\top.$$

Thus $\Pi - F\Pi F^\top \succ 0$; conjugation by $\Pi^{-1/2}$ shows that $\rho(F) < 1$. Initialize $S_1 \sim \mathcal{N}(0, \Pi)$. Induction in [Eq. \(5.36\)](#) now keeps the error covariance equal to Π , proving [Eqs. \(5.22\)](#) and [\(5.23\)](#).

Finally,

$$S_t = A^{t-1}S_1 + \sum_{j=2}^t A^{t-j}V_j.$$

The geometric decay of powers of A bounds its second moment uniformly in t ; $Z_t = CS_t + W_t$ has the same property. Geometric decay of powers of F also supplies the constants G, γ in [Proposition 5.9.2](#). $\square$

5.A.2 Why the Hankel factors recover the predictor

Proposition 5.9.3 (Recovery of the prediction map). *The recovered matrices satisfy*

$$\overline{C} \overline{F}^{j-1} \overline{L} = K_j \quad \text{for every } j \geq 1.$$

Thus Eq. (5.16) computes the same predictor on every observed history.

Proof. Under the ranks assumed in Section 5.9.2, both $\mathcal{H}_0 = \mathcal{O}_\star \mathcal{R}_\star$ and $\mathcal{H}_0 = \mathcal{O} \mathcal{R}$ are full-rank factorizations through a $d_\star$ -dimensional space. Their left factors have the same column space, so $\mathcal{O} = \mathcal{O}_\star T_0$ for an invertible T_0 . Multiplying by a left inverse of $\mathcal{O}_\star$ gives $\mathcal{R} = T_0^{-1} \mathcal{R}_\star$. Consequently, Eq. (5.15) yields

$$\overline{F} = T_0^{-1} F_\star T_0, \quad \overline{C} = C_\star T_0, \quad \overline{L} = T_0^{-1} L_\star.$$

Substituting these into $\overline{C} \overline{F}^{j-1} \overline{L}$ proves Proposition 5.9.3. Small singular values make inversion sensitive to coefficient errors; the truncated SVD selects which directions to retain. $\square$

5.A.3 State-space error and long-run statistics

Consider two state processes on one probability space:

$$S_{t+1} = g(S_t) + V_{t+1}, \quad \tilde{S}_{t+1} = \hat{g}(\tilde{S}_t) + \tilde{V}_{t+1}.$$

Assume that $\hat{g}$ is ρ -Lipschitz, $0 \leq \rho < 1$, and that the local coupled RMS error is bounded:

$$\left(\mathbb{E} \left[\|g(S_t) + V_{t+1} - \hat{g}(S_t) - \tilde{V}_{t+1}\|_2^2 \right] \right)^{1/2} \leq \varepsilon.$$

With $e_t = (\mathbb{E}[\|S_t - \tilde{S}_t\|_2^2])^{1/2}$, inserting $\hat{g}(S_t) + \tilde{V}_{t+1}$ between the next states gives $e_{t+1} \leq \varepsilon + \rho e_t$. Iterating gives the same geometric-sum bound used in Lecture 4:

$$e_t \leq \rho^{t-1} e_1 + \varepsilon \frac{1 - \rho^{t-1}}{1 - \rho}.$$

For observations $Z_t = r(S_t) + W_t$ and $\tilde{Z}_t = \hat{r}(\tilde{S}_t) + \tilde{W}_t$, assume that $\hat{r}$ is L_r -Lipschitz and

$$\left(\mathbb{E} \left[\|r(S_t) + W_t - \hat{r}(S_t) - \tilde{W}_t\|_2^2 \right] \right)^{1/2} \leq b.$$

The observation discrepancy satisfies

$$d_t := (\mathbb{E}[\|Z_t - \tilde{Z}_t\|_2^2])^{1/2} \leq L_r e_t + b.$$

For an L_f -Lipschitz function f ,

$$|\mathbb{E}[f(Z_t)] - \mathbb{E}[f(\tilde{Z}_t)]| \leq L_f d_t.$$

If both observation processes have RMS norm at most B_Z , then

$$\left\| \mathbb{E}[Z_t Z_s^\top] - \mathbb{E}[\tilde{Z}_t \tilde{Z}_s^\top] \right\|_F \leq B_Z (d_t + d_s). \quad (5.38)$$

To see this, expand the matrix difference using $(Z_t - \tilde{Z}_t) Z_s^\top + \tilde{Z}_t (Z_s - \tilde{Z}_s)^\top$, and apply Cauchy–Schwarz. Uniformly small coupled error therefore controls means, Lipschitz costs, and fixed-lag second moments.

For linear Gaussian models in common state coordinates, take $V_t = \Sigma_V^{1/2} \Xi_t$ and $\tilde{V}_t = \hat{\Sigma}_V^{1/2} \Xi_t$ with shared standard Gaussian Ξ_t , and couple the observation noises similarly. If the true states have RMS norm at most B_S , sufficient local bounds are

$$\begin{aligned}\varepsilon &\leq \|A - \hat{A}\|B_S + \|\Sigma_V^{1/2} - \hat{\Sigma}_V^{1/2}\|_F, \\ b &\leq \|C - \hat{C}\|B_S + \|\Sigma_W^{1/2} - \hat{\Sigma}_W^{1/2}\|_F.\end{aligned}$$

The contraction condition is $\|\hat{A}\| < 1$. A change of coordinates can express contraction in a weighted Euclidean norm. These bounds control errors in observation statistics uniformly over time.

5.A.4 Cayley–Hamilton

Theorem 5.9.6 (Cayley–Hamilton). *Let $A \in \mathbb{R}^{d \times d}$ have characteristic polynomial*

$$\chi_A(\lambda) = \det(\lambda I - A) = \lambda^d + a_{d-1}\lambda^{d-1} + \cdots + a_1\lambda + a_0.$$

Then

$$\chi_A(A) = A^d + a_{d-1}A^{d-1} + \cdots + a_1A + a_0I = 0. \quad (5.34)$$

Proof. The adjugate identity is

$$(\lambda I - A) \operatorname{adj}(\lambda I - A) = \chi_A(\lambda)I.$$

Write the adjugate, a matrix polynomial of degree at most $d-1$, as $\sum_{j=0}^{d-1} \lambda^j D_j$. Comparing coefficients yields

$$D_{d-1} = I, \quad D_{j-1} - AD_j = a_j I \quad (1 \leq j \leq d-1), \quad -AD_0 = a_0 I.$$

Multiply the middle identities on the left by A^j , add them to the last identity and to $A^d D_{d-1} = A^d$. All the D_j terms cancel, leaving $\chi_A(A) = 0$. $\square$